PTC On-board Segment– Energy Management

Interface Control Document (ICD)

WCR-ICD-1095

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1. PTC ON-BOARD SEGMENT – ENERGY MANAGEMENT (EM) MESSAGES

1.1 Scope and Introduction

This document contains messages passed between a PTC On-board Segment and an Energy Management (EM) system that operates on a locomotive. The PTC on-board segment connects to Energy Management as an EMP client and Energy Management acts as an EMP server. The port number used for the Class D link connection is 10050.

The interchange of the PTC Status Data (0500) message and EM Status (0600) message are 'heartbeat' messages between the PTC on-board segment and Energy Management. Using these messages, the PTC on-board segment and Energy Management negotiate the message "interface version" and determine which functions are supported and in operation between the two systems.

Other messages exchanged between PTC and EM support the exchange of track data information, restriction information, and feedback to the crews.

1.2 References

The following documents were used in the preparation of this document.

#	Title	Source	Revision	Date
1.	Positive Train Control Office – Locomotive Segment Interface Control Document (ICD). See Note A	Wabtec	Version 2.11.1	10/26/2012
2.	PTC Back Office - Energy Management Interface Control Document.	Wabtec	Version 1.4	11/2/2012
3.	AAR MSRP Section K-IV, Office Architecture and Railroad Electronics Messaging	Association of American Railroads		effective August 2012
4.	PTC Track Data Model: S-9501 S-9351	Association of American Railroads	Version 1	Pending
5.	I-ETMS Back Office Segment Requirements Specification	Wabtec	1.37	Pending
6.	I-ETMS On-board Data Distribution Messages Interface Control Document	Wabtec	Version 1.2	7/3/2012
7.	I-ETMS Security Implementation Guide	Wabtec	Version 1.2	2/17/2012

NOTES:

A This document corresponds to the pending release of AAR S-9352B.

1.3 Message Format

Message content is in the form of a message header and data. Every message will have an EMP header field followed by message data and a data integrity value. The message content will segment as shown in the table below.

Field	Size	Type	Description	Note
EMP Message Header	17	Message Header	Contains the information that is common to all messages sent between Energy Management and PTC. Explained below in the following section.	See section 1.4 for definition
Message Content	*	Variable	This is the message payload. See the following sections for specifics.	See section 1.6 for message descriptions
Data integrity value	4	CRC-32	This is the EMP data integrity check and is a CRC computed around the EMP Message Header + Message Content.	See section 1.7 for definition.

1.4 Edge Message Protocol Use Conventions

Each message has a 17-byte EMP header preceding the data message. The header format is:

Field	Size	Type	Description	Note
Protocol (header) Version	1	Numeric	0x04 = Version 1 of S-9354	
Message Type(ID)	2	Numeric	Message ID number (0 – 65535)	
Message Version	1	Numeric	Version number of message in range 1 – 255. This is to accommodate backward compatibility to build or revision levels.	
Flags	1	Numeric	0x09 = UTC Timestamp, no encryption, no compression, CRC. See S-9354 section 3.2.1.4	No other options allowed
Data Length	3	Numeric	Size of the payload of each message, in bytes. Does NOT include the header or the 4 bytes of the CRC-32.	
Message Number	4	Numeric	This is sequence number of the message which is incremented by one for each new transmission.	Retransmissions use same message number.
Message Time	4	Numeric	UTC Time of message. See S-9354 section 3.2.2.2 for options	
Variable Header Size	1	Numeric	0x00 = No variable header used	

NOTES:

1.5 PTC to Energy Management Messages

The following messages are sent from a PTC on-board segment to Energy Management.

Message ID	Message Name	Supported Message Versions for "Interface Version" 3 (identified in 0500/0600 message)
0500	PTC Status Data	5
0501	EM Access Screen Status	2
0504	OPK Encryption Key	1
0505	Operational Private Key	3
0506	Railroad Office Addresses	2
0510	Track Database Information	3
0511	Track Clear	4
0514	Track Database Data	3
0520	Summary Train Consist	3
0521	Locomotive Run Status	2
0523	Train and Crew Data	3
0530	Current Position	3
0531	Train Route	3
0532	Target List	4
0533	Speed Restriction List	4
0534	Restriction Data	2
0540	EM Banner Display Status	2
0541	EM Soft Key Status	2
0542	EM Control Indicator Status	3
0544	ITF/BCP Overlay Status	1

1.5.1 L2I218 PTC Status Data (0500) - Version 5

This message conveys status of PTC data to Energy Management. PTC and Energy Management negotiate and reach agreement on the "Interface version" (achieved through 0500-0600 message exchanges) before PTC sends any other types of messages to Energy Management. The fields "Summary train consist CRC", "Subdivision ID", "Subdivision version", "Subdivision CRC", "Restriction Aggregate CRC", and "Locomotive run status ID" must match with the EM data for these fields, or else PTC and Energy Management are out-of-synchronization (OOS). When OOS Energy Management needs to obtain the data needed to become synchronized with PTC for consistent operation of the two systems. Energy Management is responsible for determining the correct operating behavior when it is OOS with PTC.

	Field	Size (Bytes)	Type	Description	Note
1	Interface version	1	Numeric	0x04	Note A
2	PTC state	1	Enumeration	0 – Undetermined 1 – Self Test 2 – Initializing 3 – Disengaged 4 – Cut Out 5 – Active 6 – Restricted 7 – Failed	Note B
3	PTC additional status 1	1	HGFEDCBA	Additional PTC operating state information	Note C
4	PTC additional status 2	1	HGFEDCBA	PTC substate information	Note I
5	EM status	1	HGFEDCBA	EM status acknowledgement	Note D
6	Locomotive ID	10	Alphanumeric	ID of host locomotive (See Data Dictionary sect.)	
7	Summary train consist CRC	4	CRC-32	(See Data Dictionary sect.)	Note E
8	Track data subdivision count	1	Numeric	Number of track subdivisions to be synchronized.	Note F
Loop once for each subdivision *<Start Subdivision Information Loop>*					
8.1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
8.2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
8.3	Track Data version	4	Numeric	From corresponding Track Database (0510) message	
8.4	Track Version CRC	4	CRC-32	From corresponding Track Database (0510) message	
<End Subdivision Information Loop>					

9	Locomotive run status ID	1	Numeric	Locomotive run status ID number (see message 521)	Note H
10	Restriction Aggregate CRC	4	CRC-32	Speed restriction aggregate CRC for determining current applicable restrictions.	Note G

NOTES:

A. PTC initially reports the current EM interface version

B. Enumeration value "Undetermined" is specified if PTC state cannot be obtained or determined.

C. PTC Additional Status 1

a. Bit A: 1 = PTC Penalty Brake Cut Out Switch is Cut Out
b. Bit B: 1 = PTC is not receiving valid cab signal indication or cab signal system is Cut Out or Partial Cut Out
c. Bit C: 1 = Departure test is required in order for PTC to become ACTIVE
d. Bit D: 1 = PTC on-board segment is out of communication with PTC office segment
e. Bit E: 1 = Parking function is enabled
f. Bit F: 1 = PTC datasets are out of sync with the office for the current subdivision/district that the train occupies; value is 1 if the position is unknown.
g. Bit G: 1 = The locomotive air brake system is Cut Out
h. Bit H: 1 = PTC is operating under Restricted Speed rules

D. EM Status

a. Bit A: Spare
b. Bit B: 1 = Energy Management Advisory Functions Ready Acknowledge
c. Bit C: 1 = Energy Management Cruise Functions Ready Acknowledge
d. Bit D: 1 = Energy Management Cruise Functions Active Acknowledge
e. Bit E: Spare
f. Bit F: Spare
g. Bit G: Spare
h. Bit H: Spare

E. This CRC value is the one that is sent in a Summary Train Consist (0520) message.

F. Entries 8.1 through 8.4 are repeated for each track data subdivision (entry 8).

G. PTC includes a Restriction Aggregate CRC value for the current set of restrictions, which is an indicator to EM as to whether EM has the most current set of speed restrictions, i.e. Speed Restriction List (0533) and Speed Restriction Data (0534) message information. If EM does not have a match on the Restriction Aggregate CRC value, EM may then request the Speed Restriction List (0533) message by sending a Restriction List Request (0633) message. The Restriction Aggregate CRC is a CRC of each set of restriction data sets (listed within the Speed Restriction (0533) message) starting with the lowest Individual Restriction Reference Number to the highest Individual Restriction Reference Number. In the event there are duplicate individual restriction reference numbers (different SCAC), the order will be done alphabetically by SCAC (ex: "BNSF" > "CSX" > "UP") for the duplicates.

The table below shoes the fields over which the 'Restriction Aggregate CRC' is calculated in order.

Field	**Size (Bytes)**	**Type**
Repeated for each Individual Restriction Reference Number listed in the Speed Restriction List (0533) message. The CRC is computed on the Speed Restriction List (0533) message fields 2.1 through 2.3 in the order as they are listed within the Speed Restriction List (0533) message.		
Individual Restriction Reference Number	4	Numeric
Restriction SCAC	4	Alphanumeric
Restriction Type	1	Enumeration

H. The "Locomotive run status ID" is the one sent in the last Locomotive Run Status (0521) message

I. PTC additional status 2
   a. Bit A: 1 = BRAKING (enforcement braking is in process)
   b. Bit B: 1 = STOPPED (Note L)
   c. Bit C: 1 = EM Access Screen Active (data currently being displayed)
   d. Bit D: 1 = WARNING
   e. Bit E: Spare
   f. Bit F: Spare
   g. Bit G: Spare
   h. Bit H: Spare

J. General: Maximum elapsed time for sending a PTC Status Data (0500) message is TBC 224 seconds. An updated message will be sent immediately upon any change.

K. General: The train is considered stopped when the speed falls below TBC 163 MPH.

L. General: The combination of an Individual Restriction Reference Number and the SCAC of the track owner are required to uniquely identify an individual restriction due to the possibility of different track owners using the same restriction numbers when a train travels over track owned by multiple railroads.

M. When the crew is logged off the following values shall be set to zero:

- Summary Train Consist CRC
- Track Data Subdivision Count
- Locomotive Run Status ID
- Restriction Aggregate CRC

N. The following data is only guaranteed to be correct when PTC is in the 'DISENGAGED', 'ACTIVE' OR 'RESTRICTED' state:

- Summary Train Consist CRC
- Track Data Subdivision Count
- Locomotive Run Status ID
- Restriction Aggregate CRC

1.5.2 L2I160 EM Access Screen Status (0501) - Version 2

This message provides the key index for a softkey pressed while Energy Management Access Screens are active.

	Field	Size (Bytes)	Type	Description	Note
1	Key index	1	Numeric	0 to 7	Note A
2	Access Screen Status	1	Enumeration	0: Screen inactive 1: Screen active 2: Softkey pressed	
3	Message Coordination Number	4	Numeric	Use Coordination Number from Energy Management Access Screen (0601) message when sent as a response. Use 0x00000000 when sent unsolicited to represent a keypress or the screen activity state changes.	

NOTES:

A. Key 7 is reserved for exiting the Energy management Access Screen

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1.5.3 L2I397 OPK Encryption Key (0504) - Version 1

This message is sent from PTC to EM to provide a portion of the key needed to decrypt the HMAC OPK. EM will respond with an OPK Encryption Key Reply (0604) message containing the remainder of the information needed to decrypt the HMAC OPK.

	Field	Size (Bytes)	Type	Description	Note
1	P Value	16	Hexadecimal	Fixed at 0xD4A2559DF0A39C04C0BD16DA75D662B3	
2	G Value	1	Hexadecimal	Fixed at 0x05	
3	A Value length	1	Numeric	Size of A value in bytes,	
4	A Value	16	Hexadecimal	"A" value used for calculating "s" value	Note A
5	Message Coordination Number	4	Numeric	PTC assigned Coordination Number for use by OPK Encryption Key Reply (0604) message.	

NOTES:

A. The A value is based on the following calculation:

A = g^a mod p
Where:
a = 128 bit random number generated by PTC
g = 0x05
p = 0xD4A2559DF0A39C04C0BD16DA75D662B3

Additional information on this process will be provided in a restricted distribution white paper for those with a need to know.

1.5.4 L2I161 Operational Private Key (0505) - Version 3

This message provides the locomotive's Operational Private Key for Energy Management to use for offboard message authentication.

	Field	Size (Bytes)	Type	Description	Note
1	OPK	32	Numeric	Encrypted key value for use in offboard communications.	
2	Message Coordination Number	4	Numeric	Use Coordination Number from Operational Private Key Request (0605) message.	

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1.5.5 L2I342 Railroad Office Addresses (0506) – Version 2

This message provides the default EMP address of all railroads selected for operation during initialization along with the employing railroad for the crew and the owning railroad of the locomotive. This message is sent in response to a Railroad Address Office Request (0606) message.

	Field	Size (Bytes)	Type	Description	Note
1	Number of railroads	1	Numeric	Number of railroads	
Data for each Railroad. These fields are repeated for each Railroad. *<Beginning of Railroad Information Loop>*					
1.1	Railroad SCAC	4	Alphanumeric	Railroad SCAC	Note A
1.2	Railroad state	1	HGFEDCBA	Railroad locomotive Relationship	Note C
1.3	EMP address length	1	Numeric	Length of the EMP Address	
1.4	Default Office EMP address	Variable	Alphanumeric	Default EMP Address of the office for the railroad specified	
<End of Railroad Information Loop>					
2	AG address length	1	Numeric		
3	AG IP address	Variable	Alphanumeric	IP Address of the AG	Note B
4	AG port number	2	Numeric	Port number to connect to the AG	
5	Message Coordination Number	4	Numeric	Use Coordination Number from Railroad Office Address Request (0606) message.	

NOTES:

A. Fields 1.1 through 1.4 are repeated for each railroad selected for operation.

B. The AG address provided is for the EM to make a class D connection to the current applicable AG.

C. The Railroad State

Bit A: 1 = Owning Railroad
Bit B: 1 = Employing Railroad
Bit C: 1 = Operating Railroad
Bit D: Spare
Bit E: Spare
Bit F: Spare
Bit G: Spare
Bit H: Spare

1.5.6 L2I162 Track Database Information (0510) - Version 3

This message contains information to describe the attributes of one complete subdivision/district of EM-formatted track data. It is sent in response to receiving an EM Track Request (0612) message.

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
3	Track data version	4	Numeric	Track Data version for subdivision/district ID	Note A
4	Track version CRC	4	CRC-32	CRC-32 of track data version. Used only for version verification	Note A
5	File size	4	Numeric		Note A
6	Message Coordination Number	4	Numeric	Use Coordination Number from Track Request (0612) message.	

NOTES:

A. The fields noted by “Note A” are zeroed out if PTC is unable to obtain this information for the given Subdivision/District ID.

1.5.7 L2I163 Track Clear (0511) - Version 3

This message commands the clearing of track data for a single subdivision in anticipation of updated track data. This message is sent to EM when PTC loads a new version of subdivision track data.

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/District ID	2	Numeric	PTC Subdivision/district ID	
3	Track data version	4	Numeric	Track Data version for subdivision/district	Note A
4	Track version CRC	4	CRC-32	CRC-32 of track data version. Used only for version verification	Note A

NOTES:

A. This is the old (obsolete) subdivision version number and CRC that EM has currently. PTC uses this message to tell EM to delete this version of track data from its data store in anticipation of PTC sending a new version of this track data.

1.5.8 L2I345 Track Database Data (0514) - Version 3

This message is sent in response to receiving an EM Track Database Data Request (0614) message. This message contains the track data for a "portion" of one complete subdivision of track data. (Please see AAR document S-9501 (reference 6) for the format of a complete subdivision of EM-formatted track data.)

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/District ID	2	Numeric	PTC Subdivision/district ID	
3	Track data version	4	Numeric	Track Data version for subdivision/district	
4	Track version CRC	4	CRC-32	CRC-32 of track data version. Used only for version verification	
5	Offset (into data file)	4	Numeric	0-based offset into subdivision track data	Note A
6	Size of data transmitted	2	Numeric	Maximum size is 14000 bytes	Note B
7	Data	Variable	Binary	Sequence of bytes (a byte stream) starting at the specified 0-based "Offset" and having length that of the specified "Size of data".	
8	Data CRC-32	4	CRC-32	CRC value covering data (field 7) only	
9	Message Coordination Number	4	Numeric	Use Message Coordination Number from Track Database Data Request (0614) message.	

NOTES:

A. The 0-based "Offset" counts the beginning byte as byte 0 and is the number of bytes offset into the EM subdivision data file at which to begin collecting data for the "Data" portion of this message. The amount of data to be collected for this message, including the offset and onwards, is the value in the "Size of data" field.

B. This value corresponds to the size in bytes of the data being sent within this particular message in field 7 rather than the total file size. In the event of a file size mismatch between the stated file size and the received file size EM is to assume the file is corrupted and issue a new Track Database Data Request (0614) message.

1.5.9 L2I165 Summary Train Consist (0520) - Version 3

This message provides summary train consist information. This message uses a similar format to the "Train Consist (01030)" message of the Electronic Train Management System Office – Locomotive Segment Interface Control Document (ICD), with an additional enumeration value for "Reason for sending."

Note: A" detailed" train consist will be provided through a separate message.

	Field	Size (Bytes)	Type	Description	Note
1	Reason for sending	1	Enumeration	0 – Not used 1 – Response to EM request 2 – Unsolicited (BOS is source) 3 – Crew updated consist NOT IN CRC	Note A
2	Train length	2	Numeric	(40 - 25,000) Total train length in feet. (locomotives and cars)	
3	Lead locomotive orientation	1	Enumeration	0 – Not used 1 – Front (default) 2 – Back	
4	Number of loaded car count	2	Numeric	(0 – 999)	
5	Number of empty car count	2	Numeric	(0 – 999)	
6	Trailing Tonnage	2	Numeric	(0 – 45,000) Short tons	
7	Maximum train speed	1	Numeric	(0 – 125) MPH - Maximum train speed	
8	Train type	1	Enumeration	0 – Unknown 1 – Freight 2 – Passenger 3 – Intermodal 4 – High Speed Passenger 5 – Tilt Train 6 – Commuter	
9	Number of locomotives	1	Numeric	(1 – 24) All locomotive ID's in train	Note B
Data For Each Locomotive. These fields are repeated for each locomotive. *<Beginning of Locomotive Information Loop>*					
9.1	Locomotive ID	10	Alphanumeric	Locomotive ID (see Data Dictionary sect.)	
9.2	Locomotive position	2	Numeric	(1-999) Position of the locomotive in the train	

	Field	Size (Bytes)	Type	Description	Note
9.3	Tonnage of locomotive	2	Numeric	(50 – 400) Short tons	
9.4	Locomotive status	1	Enumeration	0 – Unknown 1 – Run 2 – Isolated	
9.5	Locomotive length	1	Numeric	(0- 255) Feet	
9.6	Locomotive horsepower	2	Numeric	(0 - 10,000) horsepower	
<End of Locomotive Information Loop>					
10	Number of operative brakes	2	Numeric	(0 – 999) operative brakes	
11	Number of axles	2	Numeric	(0 – 3996) axles	
12	Total braking force	4	Numeric	(0 – 30,000,000) lbs – Total braking force for the train excluding the locomotives listed under the 'Number of Locomotives' field above.	
13	Source of Consist Information – Railroad SCAC	4	Alphanumeric	Railroad SCAC of the office that provided summary consist information	
14	Consist Source – EMP address length	1	Numeric	Length of EMP Address	
15	Consist Source – EMP address	Variable	Alphanumeric	EMP Address of the office that provided summary consist information	
16	CRC	4	CRC-32	This is the CRC around all the fields from the 'Train length' through 'Consist Source – EMP address', fields 2 through 15	
17	Message Coordination Number	4	Numeric	Use Coordination Number from Summary Train Consist Request (0620) message. Use 0x00000000 when sent unsolicited	

NOTES:

A. The value "Response to EM request" is used when this message is sent in response to an EM Summary Train Consist Request (0620) message.

B. Entries 9.1 through 9.6 are repeated for each locomotive (entry 9).

C. The Summary Train Consist is only guaranteed to be correct when PTC is in the 'DISENGAGED', 'ACTIVE' or 'RESTRICTED' state.

1.5.10 L2I166 Locomotive Run Status (0521) - Version 2

This message provides the run/isolate status for all locomotives in the train. This message is sent when the crew updates locomotive run status, or in response to a Locomotive Run Status Request (0621) message.

	Field	Size (Bytes)	Type	Description	Note
1	Locomotive run status ID	1	Numeric	Used in heartbeat messages (500/600)	
2	Total number of locomotives in train	1	Numeric	(1 – 24) All locomotives including those in DP groups	Note A
<Start Locomotive Information Loop>					
2.1	Locomotive ID	10	Alphanumeric	Locomotive ID that is on this train including those in DP groups	
2.2	Run status	1	Enumeration	0 – unknown 1 – run 2 – isolated	
<End Locomotive Information Loop>					
3	Message Coordination Number	4	Numeric	Use Coordination Number from Locomotive Run Status Request (0621) message when responding. Use 0x00000000 when sent unsolicited	

NOTES:
A. Entries 2.1 through 2.2 are repeated for each locomotive (entry 2).

1.5.11 L2I167 Train and Crew Data (0523) - Version 3

This message provides employee ID and train IDs. It is sent whenever a train ID is established or changed, or when a crewmember successfully logs in or logs out.

	Field	Size (Bytes)	Type	Description	Note
1	Employee identifier length	1	Numeric	Length of the employee identifier	Note A
2	Employee identifier	Variable	Alphanumeric	Employee identifier	Note A
3	Employing RR SCAC	4	Alphanumeric	SCAC of the Employing Railroad	
4	Crew log in status	1	Enumeration	0 = Crew Not Logged In 1 = Crew Logged In	
5	Number of train IDs	1	Numeric		Note B
Repeat once for each Train ID *<Start Train ID Information Loop>*					
5.1	Train ID length	1	Numeric	Train ID length	Note C
5.2	Train ID	Variable	Alphanumeric	Train ID	See Data Dictionary sect.
5.3	Train ID Railroad SCAC	4	Alphanumeric	Railroad SCAC of this train ID	
<End Train ID Information Loop>					
6	Message Coordination Number	4	Numeric	Use Coordination Number from Train and Crew Data Request (0623) message when responding. PTC assigns a value when sent unsolicited.	

NOTES:

A. Employee identifier is unpopulated if the “Employee identifier length” field is valued with 0. Also, see Data Dictionary sect.

B. During initialization, the crew selects a list of train ids, one per operating railroad. Entries 5.1 through 5.3 are repeated for each train id (entry 5). If the “Number of train IDs” is a value 0, then the Train ID portion of this message is ‘empty’, which may be valid before “Initialization”, or may be invalid data if after initialization (i.e. in Disengaged or Active state). After a successful “Initialization”, the crew must have selected at least one operating railroad, and thus Train ID.

C. Train ID is unpopulated if the “Train ID length” field is valued with 0. Also, see Data Dictionary sect.

1.5.12 L2I168 Current Position (0530) - Version 3

This message provides the PTC resolution of absolute position of the train head end in terms of stored track database. Track segment offset is expressed as feet from the start (lower value milepost end) of the track segment. The Current Position (0530) message is sent every TBC 227 seconds or every TBC 228 feed of change in distance.

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/Division ID	2	Numeric	Subdivision/District ID 0 = Unknown	
3	Track segment ID	4	Numeric	0 = Unknown	
4	Offset into track segment	4	Numeric	Feet into track segment 0 = Unknown	
5	Direction of travel	1	Enumeration	0 = Unknown 1 = Increasing milepost direction 2 = Decreasing milepost direction	

NOTES:

1.5.13 L2I169 Train Route (0531) - Version 3

Track Segment-Track Segment and Track Segment-switch-Track Segment list (TBC 172 miles in front and TBC 173 miles behind the train). This is an ordered list of the track segments and switches to be encountered.

	Field	Size (Bytes)	Type	Description	Note
1	Switch Count	1	Numeric (0-254)		Note A
Complete once for each switch *<Start Switch Information Loop>*					
1.1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad owning the switch	
1.2	Subdivision ID	2	Numeric		
1.3	Switch ID	4	Numeric		
1.4	Previous Track Segment Connection	1	Enumeration		Note B
1.5	Next Track Segment Connection	1	Enumeration		Note B
1.6	Alignment	1	Enumeration		Note C
<End Switch Information Loop>					
2	Track Segment Count	1	Numeric		Note D
Complete once for each track segment *<Start Track Segment Information Loop>*					
2.1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad owning the subdivision, left justified space filled	
2.2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
2.3	Track Segment ID	4	Numeric		
2.4	Direction of Travel	1	Enumeration	0 = Unknown 1 = Increasing milepost direction 2 = Decreasing milepost direction	
2.5	Length	4	Numeric	feet	
2.6	Previous Switch Index	1	Numeric	255 if not connected to switch	
2.7	Next Switch Index	1	Numeric	255 if not connected to switch	
<End Track Segment Information Loop>					

NOTES:

A. The switch fields (1.1 – 1.6) are repeated the number of times contained in 'Switch count' (entry 1).

B. Indicates what leg of the switch the previous or next Track Segment is connected to:
   a. 0 = Facing Normal
   b. 1 = Trailing Normal
   c. 2 = Trailing Reverse
   d. 3 = Unknown

C. Indicates switch alignment
   a. 0 = Unknown
   b. 1 = Normal
   c. 2 = Reverse

D. Entries 2.1 through 2.7 are repeated for each track segment (entry 2).

E. When the on-board segment has a known locomotive track location and direction of travel, the on-board segment sends a Train Route (0531) message with the calculated route information to Energy Management at a minimum of once every TBC229 seconds or immediately upon the change of any calculated route information.

1.5.14 L2I170 Target List (0532) - Version 4

This message contains the list of active targets that have at least one end point along the current route.

Subdivision/Track Segment/Offset triplet of 0/0/0 indicates an unbounded target. If the start is unbounded, the target begins at the beginning of route. If the end is unbounded, the target ends at the end of route.

	Field	Size (Bytes)	Type	Description	Note
1	Target list ID	1	Numeric	Identifies the list of targets.	Note A
2	Target count	1	Numeric	Number of targets in this message	Note B
Repeat for each target *<Start Target Information Loop>*					
2.1	Target ID	4	Numeric		
2.2	Speed limit	1	Numeric	Speed in mph	Note C
2.3	Target state	1	Enumeration	1 = Active 2 = Predictive Warning 3 = Reactive Warning 4 = Predictive Braking 5 = Reactive Braking 6 = Predictive Emergency Braking	Note D
2.4	Start Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2.5	Start location subdivision/district ID	2	Numeric	PTC Subdivision/District ID	Note E
2.6	Start location track segment ID	4	Numeric		
2.7	Start location offset	4	Numeric		
2.8	End Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2.9	End location subdivision/district ID	2	Numeric	PTC Subdivision/District ID	Note E
2.10	End location track segment ID	4	Numeric		
2.11	End location offset	4	Numeric		

2.12	Target type	1	Enumeration		Notes G and I
2.13	Speed applicability	1	Enumeration	0 = entire train; 1 = head end only	
2.14	Restricted speed	1	Enumeration	0 = Not restricted speed 1 = Restricted speed applies	Note C
2.15	Time to enforcement	2	Numeric	Seconds until braking	Note H
2.16	SG-3 speed	1	Numeric	Speed in mph	Note J
<End Target Information Loop>					

NOTES:

A. The Target List ID is used in the Target List Acknowledge (0632) message. PTC increments the Target List ID with each modification of data. The ID is not incremented if the same data is resent due to missing acknowledgement. The value rolls over from 255 to 0.

B. The target information (2.1 – 2.15) is repeated the number of times contained in 'Target count' (entry 2).

C. If the target speed is restricted speed (entry 2.14), the 'speed limit' (entry 2.2) is the numeric value PTC uses to enforce restricted speed.

D. Target States:

a) "Active" is the default state for a target.
b) "Predictive Warning" indicates PTC is warning the crew as a result of this target based on predictive algorithms..
c) "Reactive Warning" is like Predictive Warning, but has no time to enforcement (has no countdown timer) for activating braking enforcement
d) "Predictive Braking" indicates PTC braking enforcement is occurring as a result of this target based on predictive algorithms.
e) "Reactive Braking" indicates PTC braking enforcement is occurring, but due to 'overspeed' as versus predictive algorithms.
f) "Predictive Emergency Braking" indicates PTC braking enforcement is occurring as a result of this target based on predictive algorithms

Subdivision/Track Segment/Offset triplet of 0/0/0 indicates an unbounded target.

E. If the start is unbounded, the target begins at the beginning of route. If the end is unbounded, the target ends at the end of route.

F. General: The Target List (0532) message is sent every TBC 230 seconds or as soon as a change is detected. If a single target is added, removed or modified then the whole list will be resent.

G. The target type will be populated using PTC specific enumeration values. Future additions will be added when needed and all concerned parties notified. Addition of placeholders is to minimize future updates to this message.

Enumeration Value	Description
0	• Track based speed restriction – maximum authorized speed • Speed restriction of type 'speed restriction' received in movement authority dataset
1	• Track based speed restriction – permanent speed restriction, turnout speed restriction
2	• Speed restriction received in bulletin dataset of type 'speed restriction'
3	• End of form based authority on track where form based authority is required • Stop short • Authority required to pass signal at stop on track where signal and form-based authority is required
4	• Signal indication in SG-1 or SG-2 • Absolute or intermediate signal with unknown indication • Dark absolute or intermediate signal *Note: SG-3 signal indications have enumeration value '28'* *Note: SG-4 signal indications have enumeration value '27'*
5	• Improper switch alignment
6	• Unknown switch position – not protected by signal or track circuit
7	• Unacknowledged work zone limit • Acknowledged work zone limits • Work zone train orientation change
8	• Equipment speed restriction
9	• Train ahead violation • Violation no longer being reported *Note: 'Violation no longer being reported' refers to any reason for removing a violation other than a positive clearing of that violation: timeout, loss of location on the part of the violator, or change of system state on the part of the violator.*
10	• Reserved
11	• Conditional movement authority
12	• Track circuit condition detected • Track circuit status unknown
13	• Reserved

14	• Route based cab signal except SG-3 and SG-4 *Note: SG-3 signal indications have enumeration value '28'* *Note: SG-4 signal indications have enumeration value '27'*
15	• Crew verification prior to changing track
16	• Critical alert
17	• Reverse move in form based authority
18	• Track out of service
19	• Bulletined crossing warning system malfunction
20	• Joint movement authority *Note: this enumeration covers both stop and restricted speed targets*
21	• Subdivision/district not synchronized • Cab signal cut out • Unknown location • Position uncertainty • Unknown direction
22	• Restricted
23	• Reserved
24	• Yard limits
25	• Enter main track
26	• Unknown switch position – protected by signal or track circuit
27	• SG-4 signal indication
28	• SG-3 signal indication
29	• Location Unknown Location
30	• Location Position Uncertainty Maximum
31	• Location Loss of Direction
32	• Cab Signal Cut Out
33	• Pass Signal at Stop
34	• Test Target
35	• TWE Territory
36	• Unhealthy Wireless Crossing
37	• Delayed in Block
38	• End of Form Based Authority on Signaled Track
39	• Speed enforcement cab signal
40	• Clear to next interlocking

41	• Future Addition 1 (To be implemented as needed)
42	• Future Addition 2 (To be implemented as needed)
43	• Future Addition 3 (To be implemented as needed)
44	• Future Addition 4 (To be implemented as needed)
45	• Future Addition 5 (To be implemented as needed)
46	• Future Addition 6 (To be implemented as needed)
47	• Future Addition 7 (To be implemented as needed)
48	• Future Addition 8 (To be implemented as needed)
49	• Future Addition 9 (To be implemented as needed)
50	• Future Addition 10 (To be implemented as needed)

H. This field is 0 when the target state is 'Reactive Warning' or any of the various braking target states.

I. Target types 'SG-4 signal indication' and 'SG-3 signal indication' represent track segments which are within an area controlled by a signal indicating an I-ETMS signal group 4 (SG-4) or signal group 3 (SG-3) respectively. The target speed values accompanying these two target types will be enforced but may not represent the owning railroad's operating rules. The EM system is responsible for determining the actual speed limit to apply to these target types in order to operate the train within the railroad's operating rules. The EM vendor must work directly with the individual railroads to define their respective operating rules for handling 'SG-4 signal indication' and 'SG-3 signal indication' target types.

J. For a target of type 'SG-3 signal indication, the 'SG-3 speed' field identifies the 'soft' target speed configured for the operating road where the signal is located and the current train type. For a target of any other type, this field is set to zero.

1.5.15 L2I171 Speed Restriction List (0533) - Version 4

This message is sent in response to a Restriction List Request (0633) message but can be sent unsolicited in the event of a bulletin or authority change in one of the subdivisions reported within the current PTC Status Data (0500) message. I-ETMS On-board obtains the speed restriction information initially from the office in the Bulletin Dataset (01041) and Movement Authority Dataset (01051) messages. All speed restrictions applicable to the current train type are included (not just the most restrictive). The EM provider must be capable of determining and using the most restrictive applicable speed restrictions provided. As all bulletins received from the office applicable to the current train type are passed onto EM, the EM provider must be capable of determining which bulletins are applicable and in effect for the current route.

	Field	Size (Bytes)	Type	Description	Note
1	Restriction Aggregate CRC	4	CRC-32	Overall Speed Restriction Data CRC value used for synchronization	Note A
2	Restriction Count	1	Numeric	Currently, 0-255 (zeroed out if no restrictions)	Note B
Data for each Speed Restriction. These fields are repeated for each restriction *<Beginning of Restriction Information Loop>*					
2.1	Individual Restriction Reference Number	4	Numeric	Restriction reference number assigned to each individual restriction.	
2.2	Restriction SCAC	4	Alphanumeric	SCAC of railroad owning the track restriction is on.	Note E
2.3	Individual Restriction Type	1	Enumeration	Type of restriction	Note C
<End of Time Restriction Information Loop>					
3	Message Coordination Number	4	Numeric	Use Message Coordination Number from Restriction List Request (0633) message. If sent unsolicited use 0x00000000	

NOTES:

A. The Restriction Aggregate CRC value is used to synchronize speed restriction data for each subdivision reported within the current PTC Status Data (0500) message. (See PTC Status Data (0500) and Energy Management Status (0600) messages for subdivision/district synchronization information.) PTC includes a Restriction Aggregate CRC for the current set of restrictions, which is an indicator to EM as to whether EM has the most current set of speed restrictions, i.e. Speed Restriction List (0533) and Speed Restriction Data (0534) message information. If EM does not have a match on the Restriction Aggregate CRC, EM may then request the Speed Restriction List (0533) message by sending a Restriction List Request (0633) message. See PTC Status Data (0500) message note G for a definition of the fields covered by this CRC value.

B. Entries 2.1 through 2.3 are repeated for each restriction (entry 2).

C. Note that enumeration values 1-4 are associated with information received in an office Movement Authority Dataset (01051) message, while 5-10 are associated with information received in an office Bulletin Dataset (01041) message.

0 = Not Used
1 = Track Warrant / Track Authority
2 = Track and Time / Track Permit
3 = Enter Main Track
4 = Pass Signal at Stop
5 = Temporary Speed Restriction
6 = Work Zone
7 = Grade Crossing
8 = Track Out-Of-Service
9 = Advisory / Cautionary
10 = Critical Alert
11 = Reserved

D. General: If any restrictions change the entire Restrictions List will be resent.

E. The SCAC of the subdivision/district is needed as restriction numbers could be repeated if the train travels on track owned by multiple railroads.

1.5.16 L2I470 Speed Restriction Data (0534) - Version 2

This message provides a single speed restriction applicable to this train (this train type). A separate message is sent for each bulletin or authority that contains a speed restriction. The Speed Restriction Data (0534) message is sent in response to a Speed Restriction Data Request (0634) message for a single restriction. Speed Restriction Data is requested individually from information contained within the Speed Restriction List (0533) message.

	Field	Size (Bytes)	Type	Description	Note
1	Individual Restriction Reference Number	4	Numeric	Assigned number used for tracking individual restrictions	Note A
2	Restriction SCAC	4	Alphanumeric	SCAC of railroad owning the track restriction is on.	
3	Restriction Type	1	Enumeration	Bulletin or Authority type	Note B
4	Head End Only Speed Restriction	1	Enumeration	1 = Yes 2 = No	
5	Restricted Speed	1	Enumeration	1 = Yes 2 = No	Note C
6	Restriction start time	7	Time	Restriction Start Date and Time YYYYMMDDHHMMSS (For Restriction Types 5-10 only; zero-filled otherwise)	See Data Dictionary sect.
7	Restriction end time	7	Time	Restriction End Date and Time YYYYMMDDHHMMSS (For Restriction Types 5-10 only; zero-filled otherwise)	See Data Dictionary sect.
8	Number of Time of Day Filters	1	Numeric	Number of restriction time ranges 0-7 (For Restriction Types 5-10 only; 0 otherwise	Note D
Data for each Time of Day Filter. Loop once for each filter. *<Beginning of Time of Day Filter Loop>*					
8.1	Day of Week	1	Enumeration	1 – Monday 2 – Tuesday 3 – Wednesday 4 – Thursday 5 – Friday 6 – Saturday 7 – Sunday	
8.2	Start Time Stamp	3		HHMMSS	

8.3	End Time Stamp	3		HHMMSS: Byte 0 Hour 00-23 Byte 1 Minute 00-59 Byte 2 Second 00-59	
<End of Time of Day Filter Loop>					
9	Number of Speed Restriction Segments	1	Numeric	Quantity of speed restrictions associated with this Individual Restriction Reference Number	Note F
Data for each Speed Restriction Segment. Loop once for each restriction value. *<Beginning of Speed Restriction Segments Loop>*					
9.1	Starting Track Segment ID	4	Numeric	Restriction starting segment	
9.2	Starting Offset	4	Numeric	Restriction starting segment offset	
9.3	Milepost Direction	1	Enumeration	Direction of travel between the start segment and end segment pair 1 = Increasing milepost direction 2 = Decreasing milepost direction	Note G
9.4	Ending Track Segment ID	4	Numeric	Restriction ending segment	
9.5	Ending Offset	4	Numeric	Restriction ending segment offset	
9.6	Subdivision/District ID	2	Numeric	Subdivision/district ID containing the restriction for this loop	
9.7	Speed	1	Numeric	Maximum authorized speed for restriction in mph 0xFF sent if no speed value is associated with this restriction	
9.8	Enforcement Direction	1	Enumeration	1 – Enforceable in both directions 2 – Enforceable in the increasing MP direction 3 – Enforceable in the decreasing MP direction	
<End of Time of Speed Restriction Segments Loop>					
10	DOT ID	7	Alphanumeric	DOT number of public crossing	Note H
11	Number of "ALL" Track Speed Restriction Segments	1	Numeric	Number of "ALL" track speed restrictions associated with this individual restriction reference number	Note I
<Start All Track Flag Loop>					
11.1	Start Milepost	4	Numeric	From location milepost xxxx.xxxx X 10000	
11.2	Start Milepost Modifier Type	1	Enumeration	0 – No modifier 1 – Prefix 2 – Suffix	
11.3	Start Milepost Modifier	5	ASCII	Milepost prefix or suffix	
11.4	End Milepost	4	Numeric	To location milepost xxxx.xxxx X 10000	
11.5	End Milepost Modifier Type	1	Enumeration	0 – No modifier	

				1 – Prefix 2 – Suffix	
11.6	End Milepost Modifier	5	ASCII	Milepost prefix or suffix	
11.7	Subdivision/District ID	2	Numeric	Subdivision/district ID containing the restriction for this loop	
<End All Track Flag Loop>					
12	Message Coordination Number	4	Numeric	Use Message Coordination Number from Restriction List Request (0634) message.	

NOTES:

A. The Individual Restriction Reference Number is used to coordinate speed restriction data with the Speed Restriction List. The Individual Restriction Reference Number is used to identify an individual restriction within the speed restriction list.

B. Note that enumeration values 1-4 are associated with information received in an office Movement Authority Dataset (01051) message, while 5-10 are associated with information received in an office Bulletin Dataset (01041) message.

0 = Not Used
1 = Track Warrant / Track Authority
2 = Track and Time / Track Permit
3 = Enter Main Track
4 = Pass Signal at Stop
5 = Temporary Speed Restriction
6 = Work Zone
7 = Grade Crossing
8 = Track Out-Of-Service
9 = Advisory / Cautionary
10 = Critical Alert
11 = Reserved

C. If the target speed is restricted speed, the 'speed limit' is the numeric value PTC uses to enforce restricted speed.

D. If the "Number of Time of Day Filters" value is zero then the restriction(s) apply to every day and 24 hours each day; fields 8.1 through 8.3 will be omitted as they are then unnecessary. The "Restriction Start Time" and "Restriction Stop Time" define the applicable daytime hours of the restriction for the stated day(s) of the week, i.e. from start time until stop time. The "Day of Week" value specifies which day of the week the stated restriction is in effect for. There will be a separate "Time of Day" (field 8) loop for each day in which a restriction is in effect.

E. General: If any restriction changes the Speed Restriction List will be rebuild, a new Restriction Aggregate CRC computed and an unsolicited Speed Restriction List (0533) message will be sent. In addition, the Restriction Aggregate CRC value will be updated within the PTC Status Data (0500) message.

F. Each set of Start Segment/End Segment pairs covers a contiguous set of track segments on track where the track name is unchanged. This implies that there are no alternate paths to get from the Starting Segment to the Ending Segment. There is only one path possible. Segment pairs will never cross a subdivision boundary.

G. Milepost Direction is intended to provide an unambiguous statement of travel direction for use in determining the connectivity between the start segment and end segment pair.

H. A bulletin of type “Grade Crossing” may be defined with a DOT ID in lieu of speed restriction segments. When a bulletin is defined with speed restriction segments, the ‘DOT ID’ field is set to “0000000”. When a bulletin is defined with a DOT ID, the ‘Number of Speed Restriction Segments’ field is set to zero.

I. A bulletin may be defined with a track designation of “ALL” if either the starting milepost or ending milepost exists on every track over which the bulletin is to be applied. Otherwise, each track name must be listed in a separate segment

1.5.17 L2I172 EM Banner Display Status (0540) - Version 2

PTC sends this message to Energy Management when the EM Banner is displayed in the prompt area of the PTC display. It can be removed from the display due to becoming inactive, or be hidden by a higher priority prompt text. PTC conditionally displays Energy Management-related information. PTC retains control of the screen for priority information.

	Field	Size (Bytes)	Type	Description	Note
1	Banner Display ID	2	Numeric	Denotes the specific EM Banner Display (0640) message this message is providing display status for.	Note C
2	Banner Display acknowledge	1	Enumeration	1 = Accepted 2 = Rejected – higher priority PTC message 3 = Rejected – Display not available in current state 4 = Rejected – Other 5 = Expired	Note A
3	Banner Display Status	1	Enumeration	0 = prompt inactive 1 = active, and not visible 2 = active, and visible	Note B Note D

NOTES:

A. If the EM Banner Display information received from Energy Management in an EM Banner Display (0640) message is accepted by PTC, PTC acknowledges this by returning an EM Banner Display Status (0540) message with “accepted” populated; else, “rejected” is populated along with the reason for rejection.

B. When the EM Banner Display is active and not visible due to a higher priority prompt text being displayed, the value “1” is populated. The value “0’ indicates the EM Banner Display is not active and is not displayed. The value “2” indicates the EM Banner Display is active and is actively displayed.

C. This is an echo of the Banner Display ID field from the EM Banner Display (0640) message.

D. In the event an EM Banner Display message is pre-empted by a higher priority message the EM provider may choose to resend the EM Banner Display message or adjust the display time value.

E. When the “Time to Live” for a message has expired an EM Banner Display (0540) message will be sent with the Banner Display Acknowledge field set to “Expired”.

1.5.18 L2I173 EM Soft-Key Status (0541) - Version 2

Sent in response to an EM Soft-Key Text (0641) message or when the crew presses the corresponding soft-key.

	Field	Size (Bytes)	Type	Description	Note
1	EM Soft-Key ID	2	Numeric		Note A
2	EM Soft-Key status	1	Enumeration	0 = Soft-Key inactive 1 = Soft-Key active 2 = Soft-Key pressed	Note B
3	Message Coordination Number	4	Numeric	Use Correlation Number from EM Soft-Key Text (0641) message. Use 0x00000000 when sent unsolicited.	

NOTES:

A. From the EM Soft-Key Text (0641) message sent by Energy Management.

B. Status of “inactive” sent in response to EM Soft-Key Text (0641) message if PTC will not display the soft-key. Status of ‘active’ sent in response to EM Soft-Key Text (0641) message when PTC has displayed the button with corresponding text. Status of ‘pressed’ sent when crew has pressed the EM soft-key.

1.5.19 L2I343 EM Control Indicator Status (0542) – Version 3

PTC sends this message to Energy Management when the EM Control Indicator is displayed on the PTC display. It can be removed from the display due to becoming inactive or by EM sending an EM Control Indicator (0642) message with NULL text. This message is sent in response to an EM Control Indicator (0642) message.

	Field	Size (Bytes)	Type	Description	Note
1	EM Indicator ID	2	Numeric	Indicates which EM Control Indicator (0642) message this message is in response to.	Note A
2	EM Indicator acknowledge	1	Enumeration	1 = Accepted 2 = Rejected – higher priority PTC message 3 = Rejected – Display not available in current state 4 = Rejected – Other	Note B
3	EM Indicator Status	1	Enumeration	0 = Indicator inactive 1 = Indicator active	Note C

NOTES:

A. This is an echo of the EM Control Indicator ID field included in the EM Control Indicator (0642) message.

B. If the EM Control Indicator information received from Energy Management in an EM Control Indicator (0642) message is accepted by PTC, PTC acknowledges this by sending an EM Control Indicator Status (0542) message with “accepted” populated; else, “rejected” is populated.

C. The value “0’ indicates the EM Control Indicator is not active and is not displayed. The value “1” indicates the indicator is active and is displayed.

1.5.20 L2I488 ITF/BCP Overlay Status (0544) – Version 1

This message provides the status of the current ITF/BCP Overlay display. This message is sent in response to an ITF/BCP Overlay Display (0644) message. This message is also sent whenever there is a change in the current ITF/BCP Overlay Display's status.

	Field	Size (Bytes)	Type	Description	Note
1	ITF/BCP Overlay ID	1	Numeric	ITF/BCP Overlay ID from ITF/BCP Overlay Display (0644) message this message is responding to.	Note C
2	Overlay Acknowledge	1	Enumeration	0 = Not used 1 = Accepted 2 = Rejected – Invalid Message 3 = Rejected – Other 4 = Message Expired	Note A
3	Overlay Status	1	Enumeration	0 = Prompt inactive 1 = Active – Not visible 2 = Active – Visible	Note B

NOTES:

A. If the ITF/BCP Overlay Display information received from Energy Management in an ITF/BCP Overlay Display (0644) message is accepted by PTC, PTC acknowledges this by returning an ITF/BCP Overlay Status (0544) message with "accepted" populated. If the message is improperly created or received "Rejected – Invalid Message" is populated. When the Time to Live expires PTC will remove the overlay message and send an ITF/BCP Overlay Status (0544) with "Message Expired" populated.

B. The value "0" indicates the ITF/BCP Overlay Display is not active and no valid overlay message is present to be displayed. The value "1" indicates the ITF/BCP Overlay Display is active but the current overlay message is not visible due to another activity utilizing the display. The value "2" indicates the ITF/BCP Overlay Display is active and is actively displaying the current message.

C. This is an echo of the ITF/BCP Overlay ID field from the ITF/BCP Overlay Display (0644) message.

1.6 Energy Management to PTC Messages

The following messages are sent from Energy Management to PTC.

Note that Energy Management Advisory messages contain data that is presented to the crew and PTC may discard those messages if PTC functions take priority or if it is determined by the PTC system that data from those messages cannot be supported.

Message ID	**Message Name**	**Supported Message Versions for "Interface Version" 3** (identified in 0500/0600 message)
0600	Energy Management Status	3
0601	Energy Management Access Screen	3
0604	OPK Encryption Key Reply	1
0605	Operational Private Key Request	2
0606	Railroad Office Address Request	2
0612	Track Request	3
0614	Track Database Data Request	3
0620	Summary Train Consist Request	2
0621	Locomotive Run Status Request	2
0623	Train and Crew Data Request	2
0624	Train and Crew Data Acknowledge	2
0625	In-Train Forces	1
0627	Speed Prediction	3
0628	In-Train Brake Cylinder Pressures	1
0632	Target List Acknowledge	1
0633	Restriction List Request	4
0634	Restriction Request	1
0640	EM Banner Display	2
0641	EM Soft-Key Text	2
0642	EM Control Indicator	3
0644	ITF/BCP Overlay Display	1

1.6.1 L2I174 Energy Management Status (0600) - Version 3

This message acts as a heartbeat message between Energy Management and PTC. Energy Management reports the software interface version, the Restriction Aggregate CRC and the ID, Version and CRC of all subdivisions back to PTC in this message. The heartbeat message is to be sent in response to each PTC Status Data (0500) message and upon a change of any reported data.

	Field	Size (Bytes)	Type	Description	Note
1	Interface version	1	Numeric	(see message 0500)	
2	Function support	1	HGFEDCBA	Defines readiness of EM functions	Note A
3	Operating state	1	HGFEDCBA	Defines operating status of EM functions	Note B
4	Summary Train Consist CRC	4	Numeric	CRC value from current summary train consist	
5	Locomotive run status ID	1	Numeric	Locomotive run status ID (see message 521)	
6	Restriction Aggregate CRC	4	CRC-32	Restriction synchronization value	
7	Track data subdivision count	1	Numeric	Number of track subdivisions to be synchronized	Note C
Repeat for each subdivision *<Start Subdivision Information Loop>*					
7.1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
7.2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
7.3	Subdivision/District Track version ID	4	Numeric	Track Data version for subdivision/district	
7.4	Track Version CRC	4	CRC-32	Used in conjunction with field 7.3 to uniquely determine a Subdivision/District file	
<End Subdivision Information Loop>					

NOTES:

A. Function support
- a. Bit A: Spare
- b. Bit B: 1 = Energy Management Advisory functions Ready
- c. Bit C: 1 = Energy Management Cruise functions Ready
- d. Bit D Spare
- e. Bit E Spare
- f. Bit F: Spare
- g. Bit G: Spare
- h. Bit H: Spare

B. Operating state
   a. Bit A: Spare
   b. Bit B: 1 = Energy Management Advisory functions active
   c. Bit C: 1 = Energy Management Cruise functions active
   d. Bit D: 1 = Energy Management Cruise function exit initiated
   e. Bit E: Spare
   f. Bit F: Spare
   g. Bit G: Spare
   h. Bit H: Spare

C. Entries 7.1 through 7.5 are repeated for each subdivision/district (entry 7).

1.6.2 L2I175 Energy Management Access Screen (0601) - Version 3

This message provides text to PTC for display. It is sent by Energy Management when Energy Management receives an indication from PTC that the "Energy Mgmt" soft key was pressed. Energy Management can control the refresh rate of this text by periodically sending this message.

	Field	Size (Bytes)	Type	Description	Note
1	Number of keys	1	1 byte	0 to 6	Note A
Repeat for each key *<Start Key Information Loop>*					
1.1	Key location	1	Numeric	0 to 6 (0 being left-most position, 6 being right-most position)	
1.2	Key color	1	Enumeration	1: Standard color (blue) 2: Alert color (yellow) 3: Disabled (grey) 4: Menu color (green)	
1.3	Key text	variable	Alphanumeric	Up to 30 characters, inclusive of any LF characters and a null terminating character	Note B Note C
<End Key Information Loop>					
2	Text	variable	Alphanumeric	Up to 4000 characters, inclusive of any LF characters and a null terminating character	Note D
3	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in EM Access Screen Status (0501) message response.	

NOTES:

A. The maximum number of keys possibly displayed is 8 keys, numbered from 0 to 7. Key numbers 0 to 6 are for EM functionality, and key number 7 (8th key) is reserved to exit EM access screens and return to I-ETMS. Entries 1.1 through 1.3 are repeated for each key (entry 1).

B. The key text is displayed as a maximum of 3 lines, each line up to a maximum of 9 actual characters ('actual' characters means no LF and no null terminating character). A new line can be specified using the LF new line character (0x0A).

C. The CDU may display directional arrows rather than text by displaying only the following in the Key text field:

- %d = Down arrow
- %u = Up arrow
- %r = Right arrow

- %l = Left arrow

D. The default display format for this text is up to 16 lines of text, each line up to 64 actual characters ('actual' characters means no LF and no null terminating character). A new line can be specified using the LF new line character (0x0A). Attributes and position of the text can be specified using a subset of ANSI standard terminal control sequences as defined in the table below:

Escape Sequence	Description	Notes
<ESC>c	Reset device	
<ESC>[K	Erase to end of line	
<ESC>[2J	Erase screen	
<ESC>[0m	Reset attributes	
<ESC>[{row};{column}H	Cursor home	
<ESC>[{rows}A	Cursor up	
<ESC>[{rows}B	Cursor down	
<ESC>[{rows}C	Cursor right	
<ESC>[{rows}D	Cursor left	
<ESC>[1m	Set bright attribute	
<ESC>[2m	Set dim attribute	
<ESC>[22m	Set default brightness (non-standard code)	
<ESC>[{color_idx}m	Set color	Note E
<ESC>)	Set bold font (non-standard)	
<ESC>(	Set default with respect to bold (non-standard)	
<ESC>[3m	Set italic font (non-standard)	
<ESC>[23m	Set default with respect to italic (non-standard)	

E.

Color Attribute	Color
30	Foreground Black
31	Foreground Red
32	Foreground Green
33	Foreground Yellow
34	Foreground Blue
35	Foreground Magenta
36	Foreground Cyan
37	Foreground White
40	Background Black
41	Background Red
42	Background Green
43	Background Yellow
44	Background Blue
45	Background Magenta
46	Background Cyan
47	Background White

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1.6.3 L2I398 OPK Encryption Key Reply (0604) - Version 1

This message is sent from EM to PTC to provide the remaining portion of the key needed to decrypt the HMAC OPK.

	Field	Size (Bytes)	Type	Description	Note
1	B Value length	1	Numeric	Size of the B value in bytes	
2	B value	16	Hexadecimal	"B" value used in calculation of "s" value	Note A
3	Message Coordination Number	4	Numeric	Use Coordination Number from OPK Encryption Key (0504) message.	

NOTES:

A. The B value is based on the following calculation:

B = g^b mod p
Where:
b = 128 bit random number generated by EM
g = value supplied in OPK Encryption Key (0504) message
p = value supplied in OPK Encryption Key (0504) message

Additional information on this process will be provided in a restricted distribution white paper for those with a need to know.

1.6.4 L2I346 Operational Private Key Request (0605) - Version 2

This message is a request for the locomotive's Operation Private Key. PTC will respond with an Operational Private Key (0505) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in PTC Operational Private Key (0505) message response	

NOTES:

1.6.5 L2I344 Railroad Office Address Request (0606) – Version 2

This message is a request for the Railroad Office Address data. PTC will respond with a Railroad Office Address (0506) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in Railroad Office Addresses (0506) message response.	

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1.6.6 L2I178 Track Request (0612) - Version 3

This message requests track data for a single subdivision/district. PTC will respond with a Track Database Information (0510) message.

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
3	Message Coordination Number	4	Numeric	EM assigned Coordination Number for Track Database Information (0510) message response.	

NOTES:

1.6.7 L2I347 Track Database Data Request (0614) – Version 3

This message requests a "portion" of one subdivision/district of track data (See AAR document 9501 for the layout of one complete subdivision/district of track data.) The requested data is sent back in a Track Database Data (0514) response message.

	Field	Size (Bytes)	Type	Description	Note
1	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad providing the list of subdivisions/districts, left justified space filled	
2	Subdivision/District ID	2	Numeric	PTC Subdivision/District ID	
3	Track Data version	4	Numeric	Track Data version for subdivision/district ID	
4	Track version CRC	4	CRC-32	CRC-32 of track data version	
5	Offset	4	Numeric	0-based offset into subdivision track data	Note A
6	Size of data	2	Numeric	maximum is 14000 bytes	
7	Message Coordination Number	4	Numeric	EM assigned Coordination Number for Track Database Data (0514) message response.	

NOTES:

A. The 0-based "Offset" counts the beginning byte as byte 0 and is the number of bytes offset into the EM subdivision data file at which to begin collecting data to fulfill this request. The amount of data requested to be collected by this message, including the offset byte, is the value in the "Size of data" field.

1.6.8 L2I348 Summary Train Consist Request (0620) - Version 2

This message requests current summary train consist (0520). PTC will respond with a Summary Train Consist (0520) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in Summary Train Consist (0520) message response.	

NOTES:

1.6.9 L2I181 Locomotive Run Status Request (0621) - Version 2

This message requests the locomotive run status. PTC will respond with a Locomotive Run Status (0521) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4		EM assigned Coordination Number for use in Locomotive Run Status (0521) message response	

NOTES:

1.6.10 L2I182 Train and Crew Data Request (0623) - Version 2

This message requests Train and Crew information. PTC will respond with a Train and Crew Data (0523) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in Train and Crew Data (0523) message response	

NOTES:

1.6.11 L2I183 Train and Crew Data Acknowledge (0624) - Version 2

This message acknowledges receipt of Train and Crew Data (message 0523).

	Field	Size (Bytes)	Type	Description	Note
1	Acknowledgement Indication	1	Enumeration	0 – Acknowledged 1 – NAK – CRC-32 failed 2 – NAK – Other reasons	
2	Message Coordination Number	4	Numeric	Use Message Coordination Number from Train and Crew Data (0523) message	

NOTES:

1.6.12 L2I184 In-Train Forces (0625) - Version 1

This message provides an array of in-train forces for graphic display. The maximum update rate on the display for in-train forces data, and the maximum rate this message is to be sent is 5Hz.

	Field	Size (Bytes)	Type	Description	Note
1	Force threshold	2	Numeric	Maximum allowed force	
2	Force offset	1	Numeric	Average distance between each force value (in feet).	Note A
3	Number of force values	2	Numeric	0-999	Note B
Repeat for each force value *<Start Force Values Information Loop>*					
3.1	Force value	2	Numeric (signed)	Kip	Note C
<End Force Values Information Loop>					

NOTES:

A. The "Force offset" entry is currently not used by PTC. PTC instead divides the pixel length of the train that is displayed by the number of force values to determine the separation distance (e.g. the number of train cars) for display of the "Force values".

B. Entry 3.1 is repeated for each force (entry 3).

C. The first "Force value" corresponds to coupler force at the first locomotive in the train. Each subsequent force value is one more offset back from the front of the train

1.6.13 L2I185 Speed Prediction (0627) - Version 3

This message provides the data required to display predicted speeds or basic text cues along the map.

Note: Speed predictions or basic text cues can be displayed at fixed locations along the track map. Those data elements must not exceed 10 characters and will be located on the track map based upon subdivision, track segment and offset.

	Field	Size (Bytes)	Type	Description	Note
1	Number of entries	1	Numeric	1 to 12	Note A
Data for each Speed Prediction. These fields are repeated for each prediction. *<Beginning of Predictive Data Information Loop>*					
1.1	Size of text	1	Numeric	Number of characters in Speed/Text field to display. Maximum is 10 characters.	
1.2	Speed / Text	variable	Alphanumeric	Number or text to be displayed on track map	
1.3	Railroad SCAC	4	Alphanumeric	Railroad SCAC of the railroad subdivision/district where text is to be displayed, left justified space filled	Note B
1.4	Subdivision/District ID	2	Numeric		Note B
1.5	Track segment ID	4	Numeric		Note B
1.6	Offset into track segment	4	Numeric	Feet into track segment	Note B
1.7	Blink	1	Enumeration	0 – Don't blink on display 1 – Blink on display	Note C
<End of Predictive Data Information Loop>					

NOTES:

A. Entries 1.1 through 1.7 are repeated for each predicted speed / text (entry 1).

B. The position for each predicted speed / text item is defined with the combination of Railroad SCAC, Subdivision/District ID, Track Segment ID, and Offset into the segment.

C. Attribute to control blinking on the display. This attribute should be set to '1' when the predicted speed exceeds maximum authorized speed at the location of the predicted speed.

1.6.14 L2I186 In-Train Brake Cylinder Pressures (0628) - Version 1

This message provides an array of individual rail car and locomotive brake cylinder pressures for graphic display.

	Field	Size (Bytes)	Type	Description	Note
1	Number of pressure values	2	Numeric	0-999	Note A
Repeat for each pressure valve. *<Start Brake Cylinder Pressure Information Loop>*					
1.1	BC pressure	1	Numeric	Pressure (PSI)	Note B
<End Brake Cylinder Pressure Information Loop>					

NOTES:

A. Entry 1.1 is repeated for each pressure (entry 1).

B. The first “BC Pressure” corresponds to the brake cylinder pressure on the lead locomotive.

1.6.15 L2I188 Target List Acknowledge (0632) - Version 1

This message acknowledges receipt of Target List (message 0532).

	Field	Size (Bytes)	Type	Description	Note
1	Acknowledgement indication	1	Enumeration	0 – Acknowledged 1 – NAK – CRC-32 failed 2 – NAK – Other reasons	
2	Target list ID	1	Numeric	ID from message 0532 to be acknowledged.	

NOTES:

1.6.16 L2I189 Restriction List Request (0633) - Version 4

This message requests the restriction list for all current speed restrictions for all subdivisions currently contained within the PTC Status Data (0500) message. PTC will respond with a Speed Restriction List (0533) message.

	Field	Size (Bytes)	Type	Description	Note
1	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in the Speed Restriction List (0533) message response.	

NOTES:

1.6.17 L2I471 Restriction Request (0634) - Version 1

This message is sent from EM to I-ETMS On-board to request specific speed restriction data. Each restriction will be requested separately.

	Field	Size (Bytes)	Type	Description	Note
1	Individual Restriction Reference Number	4	Numeric	Reference number of the speed restriction requested from the Speed Restriction List	
2	Restriction SCAC	4	Alphanumeric	SCAC of district/subdivision owning railroad	Note B
3	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in the Speed Restriction Data (0534) message response.	

NOTES:

A. General: No more than one request may be active at a given time.

B. Individual Restriction Reference Number and SCAC of track owner needed to define a unique restriction.

C. General: I-ETMS will respond back with a Speed Restriction List (0533) message if an invalid restriction is requested.

1.6.18 L2I190 EM Banner Display (0640) - Version 2

This message provides the data required to display an Energy Management Banner Display to the user.

	Field	Size (Bytes)	Type	Description	Note
1	Banner Display ID	2	Numeric	Unique identifier for Banner Display	Note A
2	Priority	1	Numeric	1 = high priority 2 = normal priority	Note B
3	Audible count	1	Numeric	Number of beeps when initially displayed. 2 beeps maximum	Note M
4	Audible volume	1	Enumeration	0 – Quiet 1 – Loud	
5	Time to live	1	Numeric	Max display time in seconds	Note G
6	EM Banner Display Type	1	Enumeration	Display type 0: No banner displayed Display type 1: 2 rows of 65characters (Old Advisory Text) Display type 2: 1 row of 85 characters, 3 data groups Display type 3: 1 row of 85 characters, 4 data groups	Note C Note J Note H Note M
Delete this section for type 0 display *Start of deleted section for type 0 display*					
7	Size of EM Banner Text	1	Numeric	Size in bytes of the EM Banner text	
8	EM Banner Text	Variable	Alphanumeric	Text to be displayed in the freeform text portion of the display. Insert a mandatory LF (0x0A) character at the end of the first line on a type 1 display. Insert a %d to designate the location of a countdown timer if one is used.	Note C
9	Banner Text Background Color	1	Enumeration	0 = No banner text to be displayed 1 = Gray Background – Data/Normal 2 = Cyan Background – Instructions/Emphasis	Note F
10	Banner Countdown Timer Used	1	Enumeration	Large countdown timer in lower left corner of display 0 = Not used or displayed 1 = Countdown timer displayed	Note G
Delete this section for type 1 display *Start of deleted section for type 1 display*					

11	Data Group 1 – 2 Field Label Size	1	Numeric	Size of Data Group 1 – 2 Label in bytes	
12	Data Group 1 – 2 Field Label	Variable	Alphanumeric	Name of Data Group 1 – 2 Label. Up to 5 characters maximum	
13	Data Group 1 Display Color	1	Enumeration	0 = Not Displayed 1 = Gray Background – Normal 2 = Cyan Background – Emphasis	Note E Note F Note H
14	Data Group 1 Field Label Size	1	Numeric	Size of Data Group 1 Label in bytes	
15	Data Group 1 Field Label	Variable	Alphanumeric	Name of Data Group 1 parameter to be displayed. Up to 16 characters maximum	
16	Data Group 1 Field Value Size	1	Numeric	Size of Data Group 1 Parameter Value in bytes	
17	Data Group 1 Field Value	Variable	Alphanumeric	Parameter value to be displayed. Up to 6 characters	
18	Data Group 1 Arrow Direction	1	Enumeration	0 = Not used (no arrow displayed) 1 = Arrow pointing up 2 = Arrow pointing down	Note D
19	Data Group 2 Display Color	1	Enumeration	0 = Not Displayed 1 = Gray Background – Normal 2 = Cyan Background – Emphasis	Note E Note F Note I
20	Data Group 2 Field Label Size	1	Numeric	Size of Data Group 2 Label in bytes	
21	Data Group 2 Field Label	Variable	Alphanumeric	Name of Data Group 2 parameter to be displayed. Up to 16 characters maximum	
22	Data Group 2 Field Value Size	1	Numeric	Size of Data Group 2 Parameter Value in bytes	
23	Data Group 2 Field Value	Variable	Alphanumeric	Parameter value to be displayed. Up to 6 characters	
24	Data Group 2 Arrow Direction	1	Enumeration	0 = Not used (no arrow displayed) 1 = Arrow pointing up 2 = Arrow pointing down	Note D
Start of Group 3 and 4 Information Loop *(This loop is used 1 time for Type 2 displays and 2 times for Type 3 displays)*					
25	Data Group Display Color	1	Enumeration	0 = Not Displayed 1 = Gray Background – Normal 2 = Cyan Background – Emphasis	Note L
26	Data Group Label Size	1	Numeric	Size of Data Group Label in bytes	

27	Data Group Label	Variable	Alphanumeric	Name of Data Group. Up to 5 characters maximum	
28	Data Group Field Label Size	1	Numeric	Size of Data Group Field Label in bytes	
29	Data Group Field Label	Variable	Alphanumeric	Name of the Data Group parameter to be displayed. 16 characters maximum	
30	Data Group Field Value Size	1	Numeric	Size of Data Group Parameter Value in bytes	
31	Data Group Field Value	Variable	Alphanumeric	Parameter value to be displayed. Up to 6 characters	
32	Data Group Arrow Direction	1	Enumeration	0 = Not used (no arrow displayed) 1 = Arrow pointing up 2 = Arrow pointing down	Note D
33	Data Group Fence	1	Enumeration	0 = No Data Group Fence displayed 1 = Data Group Fence displayed	
End of Group 3 and 4 Information Loop					
End of deleted section for type 1 display					
End of deleted section for type 0 display					

NOTES:

A. Energy Management is responsible for the assignment of unique Banner Display IDs. PTC will only use this identifier in EM Banner Display Status (0540) message to provide the display status on this request. Use a new Banner Display ID each time the display is updated.

B. Currently, NOT supported by PTC; all EM Display Banners are lowest priority compared to PTC prompts.
1 = High priority = Higher priority than some PTC-related prompts
2 = Low priority = Lower priority than all PTC-related prompts

C.

- To remove the EM Banner Display set size of field value to 0.
- To change any displayed information, send a new EM Banner Display (0640) message with the new information to be displayed. The current EM Banner Display will be deleted/removed (without status response) and the new EM Banner Display created.
- To extend the life of active EM Banner Display, send the same EM Banner Display (0640) message (with the same Banner Display ID) again (extending time to live).
- The priority of the EM Banner Display is the lowest with relation to PTC prompts. An active EM Banner Display is overlaid, and thus 'hidden', when any higher priority PTC Banner needs to be displayed by PTC.

D. Arrow color, when displayed, matches DP Parameter Value color.

E. Color choice covers the Data Group Field Label, Data Group Field Value and Data Group Arrow Direction fields on the display.

F. Background color used shall be consistent with ITC designated usage rules:

- Gray: Used for displaying information and optional crew actions.
- Cyan: Used for displaying changing data and required crew actions.

G. When a countdown timer is used the message time to live becomes the countdown timer value.

H. Skip fields 14 through 18 when Data Group 1 is not displayed.

I. Skip fields 20 through 24 when Data Group 2 is not displayed.

J. Skip fields 7 through 33 when a Type 0 display is selected.

K. Skip fields 11 through 33 when a Type 1 display is selected.

L. Skip fields 26 through 33 when no data group is being displayed.

M. Number of beeps allowed is limited to 2. Use a single beep for routine information changes or instructions. Use 2 beeps for priority instructions.

N. The term 'skip' used within this message means that the skipped fields are not to be populated with any values. They are to be removed from the transmitted message.

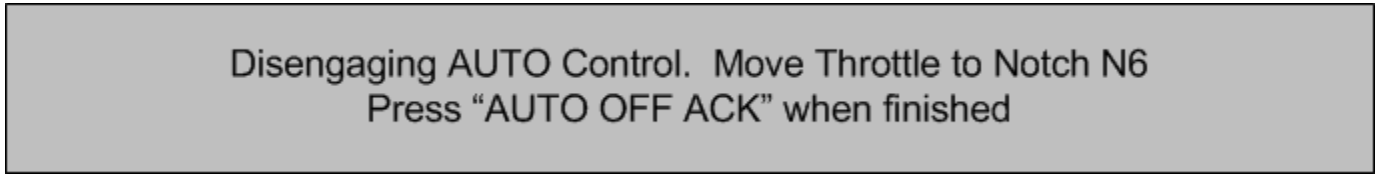

Figure 1 – Display Type 1 – 2 Line x 65 Characters

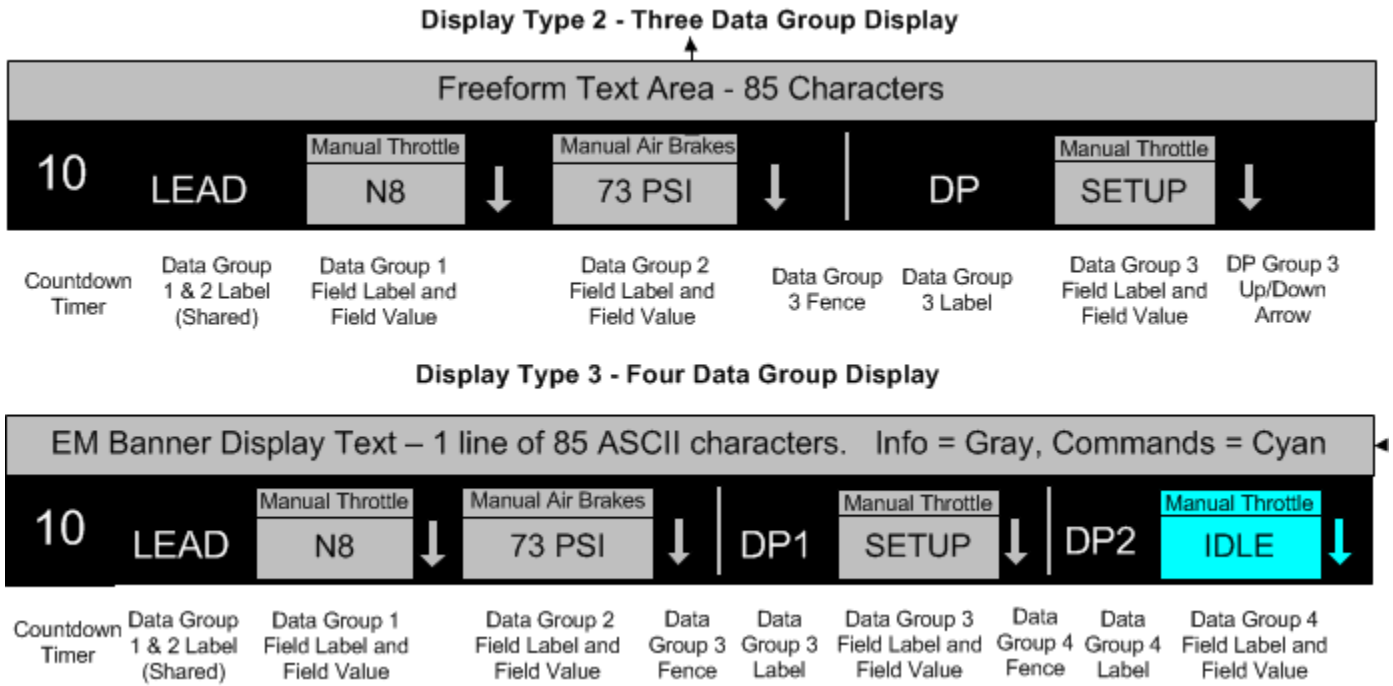

Figure 2 – Display Type 2 and Type 3 showing individual field names used in the EM Banner Display message

Note: Arrows are not labeled due to space constraints in this example.

1.6.19 L2I191 EM Soft-Key Text (0641) - Version 2

This message is sent by Energy Management to indicate when to display an Energy Management soft-key and the corresponding text. Energy Management uses this key within the context of controlling EM functions. PTC will respond with an EM Soft-Key Status (0541) message.

	Field	Size (Bytes)	Type	Description	Note
1	EM Soft-Key ID	2	Numeric	Unique identifier for soft-key	Note A
2	Key text	1	ASCII	Up to 30 characters; null terminated	Note B Note C
3	Key Color	1	Enumeration	1: Standard color (blue) 2: Alert color (yellow) 3: Disabled (grey)	
4	Message Coordination Number	4	Numeric	EM assigned Coordination Number for use in EM Soft-Key Status (0541) message response	

NOTES:

A. Energy Management is responsible for the assignment of unique EM Soft-Key IDs. PTC will use this identifier in the EM Soft-Key Status (0541) message.

B. EM Soft-Key is removed by sending a null string here.

C. EM may display up to 3 lines of 9 characters separated by a linefeed character (0x0A).

1.6.20 L2I349 EM Control Indicator (0642) - Version 3

This message provides the data required to display an Energy Management control indicator on the CDU. PTC will respond with an EM Control Indicator (0542) message.

	Field	Size (Bytes)	Type	Description	Note
1	Indicator text ID	2	Numeric	Unique identifier for text	Note A
2	Indicator text	Variable	ASCII string	Up to 16 characters; Null terminated	
3	Indicator color	1	Enumeration	0: Not used 1: Gray – Normal 2: Cyan – Emphasis	

NOTES:

A.

- Energy Management is responsible for the assignment of unique prompt IDs. PTC will only use this identifier in EM Control Indicator Status (0542) message.
- To clear the indicator display, send the message with a null string.
- To change an active indicator or change the color, send the message with the new text or color. The current indicator will be removed and the new indicator displayed.

1.6.21 L2I487 ITF/BCP Overlay Display (0644) - Version 1

This message provides the ability to display an ITF/BCP Overlay Display to be displayed over the current In-Train Forces graph and Brake Cylinder Pressure graph. This message will receive an ITF/BCP Overlay Status (0544) message in response.

	Field	Size (Bytes)	Type	Description	Note
1	ITF/BCP Overlay ID	1	Numeric	ITF/BCP Overlay ID assigned by EM	Note A
2	Type of Overlay to Display	1	Enumeration	0 = Display no ITF/BCP Overlay 1 = Display 5 Line ITF/BCP Overlay 2 = Display Top 2 Line ITF/BCP Overlay 3 = Display Bottom 2 Line ITF/BCP Overlay	Note B
3	Background Color	1	Enumeration	0 = Do not display/Cancel existing display 1 = Gray – Information/Normal 2 = Cyan – Instructions/Emphasis 3 = Yellow – Enumerated Messages	Note C
4	Message Type	1	Enumeration	0xFF = Freeform text All others = RR approved enumerated text ID for the type of display chosen in field 2	Note D
5	Size of Overlay Message	1	Numeric	Size of the overlay message in bytes	
6	Overlay Message	Variable	Alphanumeric	Information to be displayed. Insert a LF (0x0A) character between each line of the overlay message in a message type 0xFF display. 16 characters maximum per line; does not count LF character. Enumerated Message Insertions: • %d = Countdown timer, displayed as a 2 digit decimal number (counts as 2 characters) (A countdown timer can be used in Non-Enumerated messages as well) • %t = Throttle Notch, displayed as 'Nx', where x is the notch value supplied (Max value 8) (counts as 2 characters) • %b = Dynamic Braking Notch, displayed as 'Dx', where x is the braking notch supplied (Max value 8) (counts as 2 characters)	Note E Note F Note G Note H

				• %p = Braking Pressure, displayed as a 2 character decimal number (Max value 99) (counts as 2 characters) • %i = Integer Insertion, displayed as a single digit decimal number (counts as 1 character) Insertions are made at the %x location in a message, where x is the insertion character.	
7	Audible Count	1	Numeric	Number of beeps when initially displayed. Maximum of 2 beeps. Set to zero when no ITF/BCP Overlay displayed	Note I
8	Audible Volume	1	Enumeration	0 = Quiet 1 = Loud	
9	Time to Live	1	Numeric	Time to display the information on the screen in seconds	Note G Note H

NOTES:

A. Use a new Overlay Display ID each time the display is changed.

B. To remove all ITF/BCP Overlay display(s) set this field to zero. Skip fields 3 through 9.

C. Any message containing a countdown timer will be removed 1 second after the countdown timer reaches zero. At time expiration the message will be displayed with a countdown timer value of “0” for 1 second before being removed and an ITF/BCP Overlay Text Status (0544) being sent.

D. ITC maintains the list of approved enumerated messages for display. The enumerated message value chosen must be a valid choice from the list of 2 line or 5 line messages available or the result will be a rejected message in the ITF/BCP Overlay Status (0544) message response. The official ITC document will be cited when it becomes available.

E. When this is displaying an enumerated message this field will include any throttle notch value, dynamic braking notch value or brake pressure information to be inserted into the message in this field. If a countdown time is used in an Enumerated Message this field will not be populated with the timer value but will have the time to live value from field 9 inserted instead. As no Enumerated Messages have multiple insertions of a non-countdown timer parameter there is no need to have a separator at this time; however a whitespace character will be used as a separator between insertion values in the event this should be needed in the future. If no parameter is to be inserted into the Enumerated Message or only a countdown timer is inserted place a null in this field.

F. All text in both freeform text and enumerated messages is centered horizontally and vertically in the space available.

G. The ‘0’ enumeration feature for 2 line displays only: Selecting a zero here will cancel only the top or bottom 2 line display type chosen in Field 2: Type of display to overlay. Skip fields 4 through 9. The other active 2 line displays would remain being displayed if one is currently active.

H. Any countdown timer will be assigned the same value as the message’s time to live.

I. Number of beeps allowed is limited to 2. Use 1 beep for routine information changes or instructions when the operator’s attention is wanted. Use 2 beeps for Enumerated Messages or priority instructions.

J. The term ‘skip’ used within this message means that the skipped fields are not to be populated with any values. They are to be removed from the transmitted message.

1.7 Data Dictionary

When fields are comprised of multi-byte quantities the byte order is presumed to be most significant byte first; for example, if a data quantity is represented in two bytes as a binary value of 259; the byte order for this field (as byte n and n+1) as it would appear in the message byte stream is:

…..
byte n 1 (msb)
byte n+1 3 (lsb)
…..

The same convention applies to multi-byte quantities of other lengths.

Field	Size (Bytes)	Type	Description
CRC-32	4	CRC-32	The CRC-32 used is defined within S-9354. Please see this document for all details.
EM Restriction Reference Number	4	Numeric	This is special to EM and should not be confused with the 'normal' speed restriction reference number for the office 01041 and other office messages. Please see Note G for the PTC Status Data (0500) message. PTC includes an EM restriction reference number for each subdivision, which is just an indicator to EM as to whether EM has the most current set of speed restrictions, i.e. Speed Restriction List (0533) message information, for a given subdivision. If EM does not have a match on the EM restriction reference number for a given subdivision, EM may then request the Speed Restriction List (0533) message by sending a Restriction List Request (0633) message.
Interface Version	1	Numeric	This is the version of software in operation for the energy management interface, which is negotiated between PTC and Energy Management Please see Note A for the PTC Status Data (0500) message.

Field	Size (Bytes)	Type	Description
Locomotive ID	10	Alphanumeric: RR SCAC (4)+Loco No.(6)	Represents Locomotive ID as reported by locomotive to office, ASCII text comprising 4 (alpha) bytes for RR (SCAC), left justified space filled and 6 (numeric) bytes, right justified space filled for engine number. ASCII Text First 4 characters are railroad id, last six characters are unit number Ex: BNSF 208, BN 208
Time	7	Time	Byte 0-1 Year Byte 2 Month 1-12 Byte 3 Day 1-31 Byte 4 Hour 00- 23 Byte 5 Minute 00-59 Byte 6 Second 00-59 Unless otherwise noted, all time fields in this document refer to UTC time.
Train ID	Variable	Alphanumeric: RR SCAC (4)+RR assigned identifier (variable)	Represents Train ID as reported by locomotive to office, ASCII text comprising 4 (alpha) bytes for RR (SCAC), left justified space filled and variable (alphanumeric) bytes, for RR assigned identifier.
Various1	Variable	Alphanumeric	Any field of variable size and type ‘Alphanumeric’ or ASCII is defined to be a fixed length of characters.
Various2	1	HGFEDCBA	Fields broken into sizes less than 1 byte are represented with characters HGFEDCBA where H is MSB and A is LSB of the byte.

1.8 Energy Management TBC Values

Note: This TBC information is provided as reference only and is controlled within reference document 7 (see section 1.2). The information will be updated whenever the information within the controlled source is changed. In the event of a conflict in data the controlling document will take precedence over this document

TBC Number	TBC Description	Default Value	Range	Units
163	The speed at which the train is considered stopped	0.1	0 – 2.0	MPH
172	Train's Front Route Calculation Distance	8	0 - 8	Miles
173	Train's Rear Route Calculation Distance	8	0 - 8	Miles
224	Elapsed time for sending PTC Status Data (0500) message	5	5 – 5	Seconds
227	Periodic Time for sending Current Position (0530) message	5	5 - 5	Seconds
228	Distance for sending Current Position (0530) message	500	500 – 500	Feet
229	Periodic time for sending Train Route (0531) message	15	15 – 15	Seconds
230	Periodic time for sending Target List (0532) message	15	15 – 15	Seconds
231	Elapsed time for receiving EM Display Information	5	5-5	Seconds
232	Elapsed time for receiving Crew and Train Data Acknowledge (0624) message	1	1-1	Seconds
233	Elapsed time for receiving Target List Acknowledge (0632) message	1	1-1	Seconds

1.9 Definitions

1.9.1 On-Board Segment State

1.9.1.1 Self Test

A state in which the on-board segment performs a self-test of certain of its hardware and software components. The on-board segment enters this state on power-up or reboot. After a successful self test, the on-board segment transitions to CUT OUT.

1.9.1.2 Failed

A state in which a hardware or software error has been detected. The on-board segment transitions to FAILED when a failure has been detected in any other state.

1.9.1.3 Cut Out

A state in which the on-board segment has not yet initialized or has detected a condition that does not allow it to enforce. The on-board segment transitions to the CUT OUT state:

a. From SELF TEST
- After a successful self test

b. From INITIALIZING, DISENGAGED, ACTIVE or RESTRICTED
- Upon a command from a BOS
- Locomotive air brake system is cut out
- I-ETMS penalty brake cut out switch in “Cut Out” position

c. From DISENGAGED, ACTIVE or RESTRICTED
- Via crew key press to cut out or log off

d. From INITIALIZING
- After an unsuccessful initialization

e. From INITIALIZING, DISENGAGED or ACTIVE
- After an unsuccessful departure test

f. From FAILED
- When all failures are cleared in FAILED state

1.9.1.4 Initializing

A transitional state initiated by the crew in which the on-board segment validates its configuration and the crew credentials and obtains essential information about the train. The on-board segment transitions to this state from the CUT OUT, DISENGAGED and ACTIVE states via crew key press.

1.9.1.5 Disengaged

A state in which the on-board segment has initialized but is not prepared to enforce. The on-board segment transitions to this state:

a. From INITIALIZING
- After successful initialization has successfully completed

b. From ACTIVE
- When the on-board segment is not prepared to enforce

c. From RESTRICTED
- Upon crew key press

1.9.1.6 Active

A state in which the on-board segment is prepared to enforce. The on-board segment transitions to this state:

a. From DISENGAGED
- When the on-board segment is prepared to enforce

1.9.1.7 Restricted

A state in which the on-board segment enforces restricted speed but is not otherwise prepared to enforce. The on-board segment transitions to this state from the ACTIVE state via crew key press.

2. Appendix A: Enumerated Message Rules:

The following rules will be applied to the use of Enumerated Messages:

1. Each line within the Enumerated Message will be a maximum of 16 characters followed by a linefeed character to separate lines to be displayed.

2. Each Enumerated Message shall end with a NULL character.

3. The Enumerated Message will allow only predefined insertion types to be included in the displayed message. All submitted Enumerated Messages must comply with the defined format for insertable characters.

    a. When the Enumerated Message contains a '%T' the displayed Enumerated Message will insert 'NX' where 'X' is the throttle notch value provided in the request message. The '%T' counts as 2 printable characters in the character count.

    b. When the Enumerated Message contains a '%B' the displayed Enumerated Message will insert 'DX' where 'X' is the dynamic braking notch value provided in the request message. The '%B' counts as 2 printable characters in the character count.

    c. When the Enumerated Message contains a '%D' the displayed Enumerated Message will insert 'XY' where 'XY' is the base 10 value of the countdown timer value provided in the request message. The '%D' counts as 2 printable characters in the character count.

    d. When the Enumerated Message contains a '%P' the displayed Enumerated Message will insert 'XY' where 'XY' is the base 10 value of the brake pressure value provided in the request message. The '%P' counts as 2 printable characters in the character count.
    Note that there are no units or braking type associated with this brake pressure insertion. This must be included in the Enumerated Message.

4. Invalid Enumerated Message requests will generate an acknowledgement message stating an invalid display request message was sent.

5. An Enumerated Message request containing a countdown timer will have the countdown timer displayed as the message time to live value.

6. When the countdown timer within an Enumerated Message reaches zero the message will be displayed for one second with a countdown value of zero before being removed.

7. Enumerated Messages can be cancelled by sending an Enumerated Message display request with the overlay message type enumeration value set to zero or overwriting the ITF/BCP Overlay with another displayed message.

8. Enumerated Messages shall not be brand or product specific.

9. Use the term ‘AUTO’ to refer to the EM state where EM is controlling the locomotive’s throttle.

10. Use the term ‘MANUAL’ to refer to the EM state where EM is not controlling the locomotive’s throttle.